

BellaBeat Fitness Casestudy

Goodness Akinkuade

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Introduction

Company Overview

BellaBeat Fitness is a high-tech manufacturer of health-focused products for women including: smart watches, smart bracelets/neck-pieces/clips, smart water bottles, an app that provides users with health data and connects all the smart devices and a membership which allows subscribed members access to personalized reports and consultations about issues surrounding their health that includes sleep, activity/exercise, diet and calories intake, beauty and mindfulness. We've been contracted to focus on one of Bellabeat's products and analyze smart device data to gain insights into how consumers are using their smart devices. The insights will help guide marketing strategy and reveal more opportunities for growth for the company.

Stakeholders

- Urška Sršen: Bellabeat's cofounder and Chief Creative Officer.
- Sando Mur: Mathematician and Bellabeat's cofounder.
- Bellabeat marketing analytics team.

Data & Data Source

The data used for this case study is from FitBit Fitness Tracker Data, available on Kaggle and published by Mobius under a CC0 Public Domain license. The data contains fitness tracker records submitted by 30 consenting FitBit users who participated in a distributed Amazon Mechanical Turk survey conducted in 2016. Relying solely on this dataset for the case study may limit the validity and depth of the insights obtained. Therefore, it has been recommended that an additional dataset be incorporated to produce a more comprehensive and effective analysis. The dataset receives a low originality rating because it was collected and distributed by a third party. However, it is relatively comprehensive, as it includes daily, hourly, and minute-level fitness data recorded over several days for each participant. Another major limitation is its age, since the data was collected in 2016 and users' behaviors, habits, and technology usage may have changed since then. In addition, the dataset has limited source attribution, although some information about the data-collection process and the parties involved has been documented.

Data Processing

All data cleaning, formatting, manipulation, analysis and visualization have been done using R programming language in the R Studio.

```
activity <- read.csv(here("BellaBeat CaseStudy", "dailyActivity_merged.csv"))
intensities <- read.csv(here("BellaBeat CaseStudy", "hourlyIntensities_merged.csv"))
sleep <- read.csv(here("BellaBeat CaseStudy", "sleepDay_merged.csv"))
weight <- read.csv(here("BellaBeat CaseStudy", "weightLogInfo_merged.csv"))
```

```
#Viewing Datasets & Columns
```

```
head(activity)
head(intensities)
head(sleep)
head(weight)
```

```
colnames(activity)
colnames(intensities)
colnames(sleep)
colnames(weight)
```

Processing & Analysis After importing and examining the data, I observed that the date/time columns of the datasets have to be modified in order for the data to be useful for analysis.

```
#Correcting the time and date formats of the datasets.
```

```
activity$ActivityDate <- as.POSIXct(activity$ActivityDate, format = "%m/%d/%Y")
```

```
intensities <- intensities %>%
  mutate(ActivityHour = mdy_hms(ActivityHour))
```

```
sleep$SleepDay <- as.POSIXct(sleep$SleepDay, format = "%m/%d/%Y")
```

```
weight$Date <- as.POSIXct(weight$Date, format = "%m/%d/%Y %I:%M:%S %p", tz=Sys.timezone())
weight$date <- format(weight$Date, format = "%m/%d/%Y")
weight$time <- format(weight$Date, format = "%I:%M:%S %p")
```

Data Analysis

```
#Summarizing the datasets.
```

```
# activity
```

```
activity %>%
  select(TotalSteps, TotalDistance, SedentaryMinutes, Calories) %>%
  summary()
```

```
# active minutes per category
```

```
activity %>%
  select(VeryActiveMinutes, FairlyActiveMinutes, LightlyActiveMinutes) %>%
  summary()
```

```
# calories
```

```
activity %>%
  select(Calories) %>%
  summary()
```

```
# weight
```

```
weight %>%
  select(WeightKg, BMI) %>%
  summary()
```

After importing the Weight dataset, I observed a low number of unique participants, so I wrote a code to confirm.

```
#Checking participant coverage percentage of the Weight dataset, due to low number of unique users.
weight %>%
```

```
  summarise(
    unique_users = n_distinct(Id),
    expected_users = 33,
    coverage_percent = n_distinct(Id) / expected_users * 100
  )
```

```
##   unique_users expected_users coverage_percent
## 1             8             33             24.24242
```

Still using the Weight dataset, I wanted to check if there was any significant weight change noticed from the users that participated in the fitness data category.

```
#Checking to see any significant changes in weight worth reporting.
```

```
weight_change <- weight %>%
group_by(Id) %>%
  summarise(
    max_weight = max(WeightKg),
    min_weight = min(WeightKg)
  )
```

Due to the limited number of participants represented in the weight dataset and the lack of meaningful variation in the recorded weight measurements, this dataset was excluded from further analysis. As a result, no conclusions or recommendations were drawn from the weight data, as it was considered insufficient to support reliable or generalizable insights.

```
#Calculating Awake Time in Bed & Sleep Efficiency
```

```
sleep <- sleep %>%
  mutate(
    AwakeTimeInBed = TotalTimeInBed - TotalMinutesAsleep,
    SleepEfficiency = TotalMinutesAsleep / TotalTimeInBed * 100
  )
```

```
# sleep
```

```
sleep %>%
  select(TotalSleepRecords, TotalMinutesAsleep, TotalTimeInBed, AwakeTimeInBed, SleepEfficiency) %>%
  summary()
```

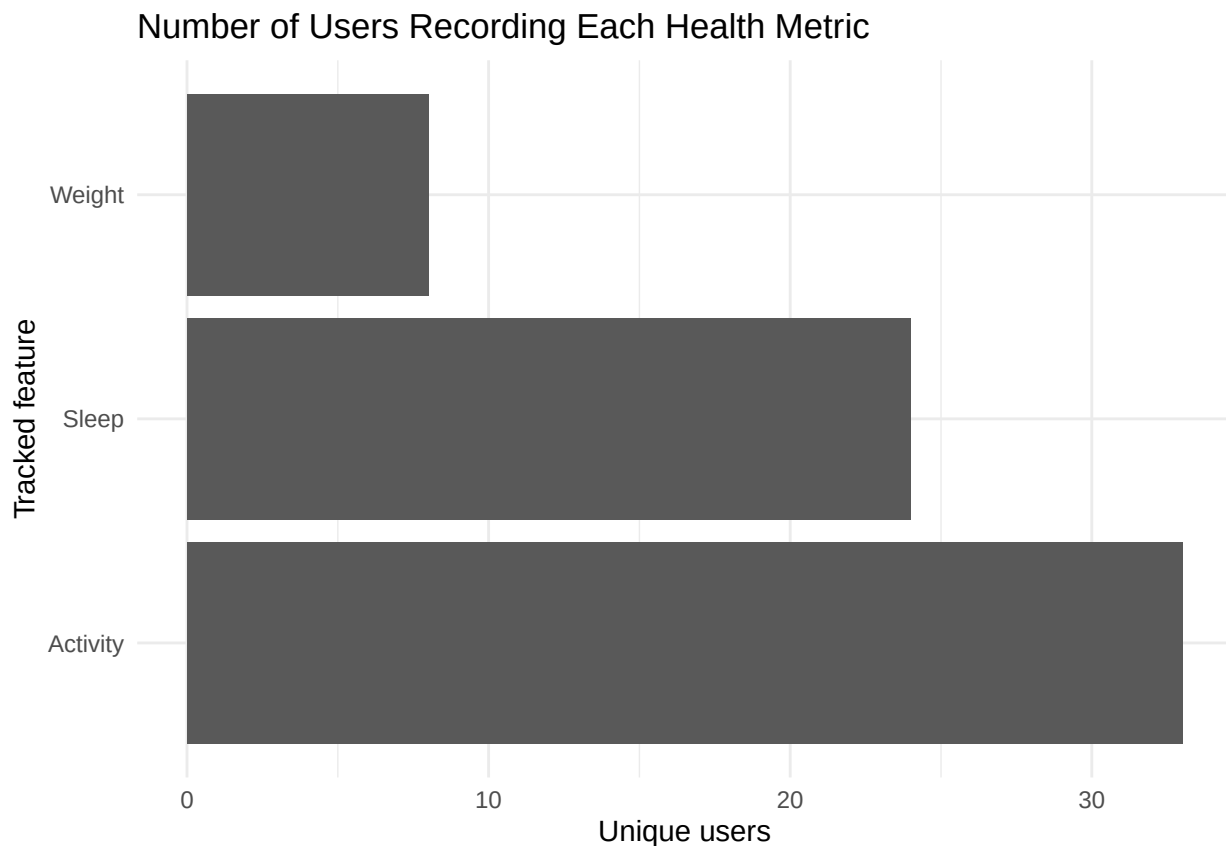
From the data and information we have for this project, I noticed that most data collected was automatically logged and not manually. This means it might not be possible to get any accurate insights about how frequently the users open the Fitbit or Bellabeat app, whether they read or follow health recommendations on the app or devices, whether certain notifications changed their behaviour or whether a specific Bellabeat product feature caused improved health. But, we can check how many unique users participated in each fitness category.

```
#Feature Usage
```

```
feature_usage <- data.frame(
  Feature = c('Activity', 'Sleep', 'Weight'),
  UniqueUsers = c(
    n_distinct(activity$Id),
    n_distinct(sleep$Id),
    n_distinct(weight$Id)
  )
)
```

```
#Viz: Number of Users Recording Each Health Metric.
```

```
ggplot(feature_usage, aes(x = Feature, y = UniqueUsers)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Number of Users Recording Each Health Metric",
    x = "Tracked feature",
    y = "Unique users"
  ) +
  theme_minimal()
```



```
#Average activity by hour of the day.
hourly_activity <- intensities %>%
  mutate(hour = hour(ActivityHour)) %>%
  group_by(hour) %>%
  summarise(
    AvgTotalIntensity = mean(TotalIntensity, na.rm = TRUE),
    AvgIntensity = mean(AverageIntensity, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(time = format(
    as.POSIXct(hour * 3600, origin = "1970-01-01", tz = "UTC"),
    "%I %p"))

#Viz: Most Active Hour of the Day
ggplot(data = hourly_activity, aes(x = hour, y = AvgTotalIntensity)) +
  geom_line() + geom_point() +
  scale_x_continuous(breaks = hourly_activity$hour,
```

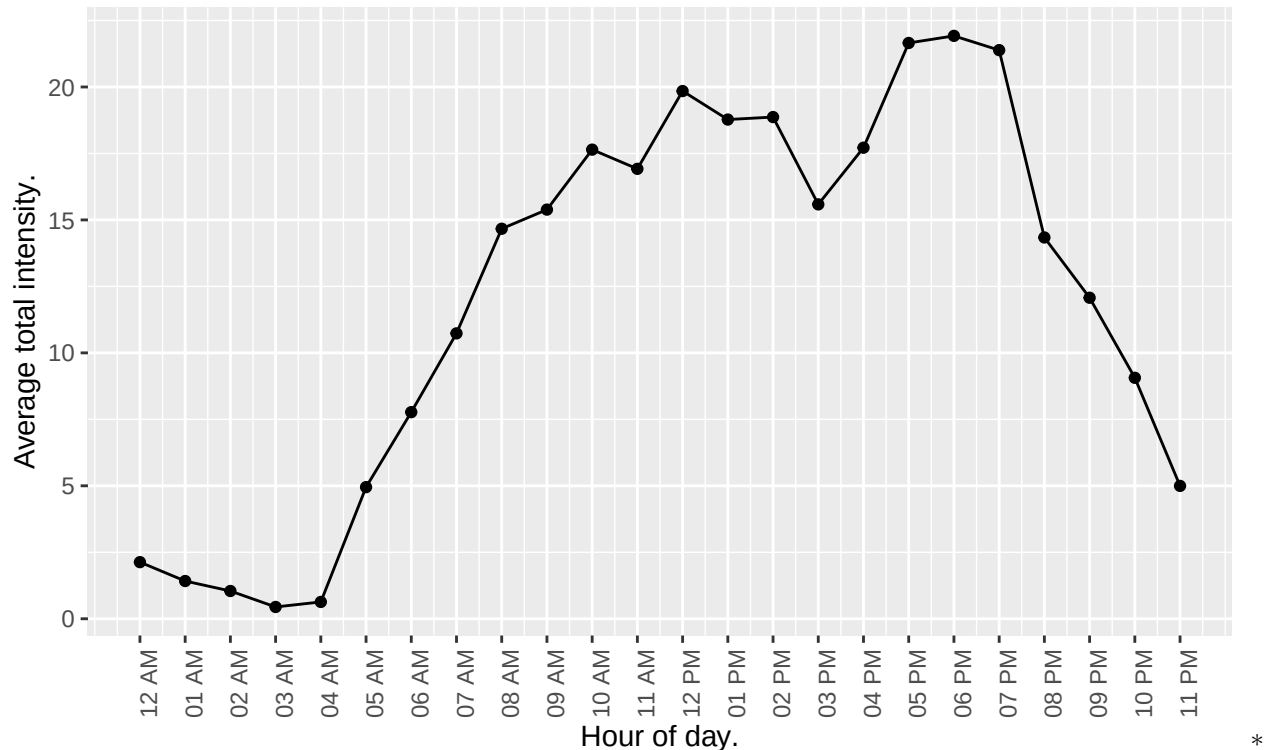
```

      labels = hourly_activity$time) +
labs(title = "Most Active Hour of the Day.",
      subtitle = "Shows the periods when users were generally most active.",
      x = "Hour of day.",
      y = "Average total intensity."
    ) +
theme(axis.text.x = element_text(angle = 90))

```

Most Active Hour of the Day.

Shows the periods when users were generally most active.



Insight: We can clearly see from the visualization that 6:00PM seems to be the most active hour of the day for the users.

```

#Average activity by day of the week.
daily_activity <- activity %>%
  mutate(day = wday(ActivityDate, label = TRUE, abbr = FALSE),
         TotalActiveMinutes = VeryActiveMinutes + FairlyActiveMinutes + LightlyActiveMinutes) %>%
  group_by(day) %>%
  summarise(AvgTotalSteps = mean(TotalSteps, na.rm = TRUE),
            AvgCalories = mean(Calories, na.rm = TRUE),
            TotalAvgActiveMinutes = mean(TotalActiveMinutes, na.rm = TRUE),
            .groups = "drop"
  )

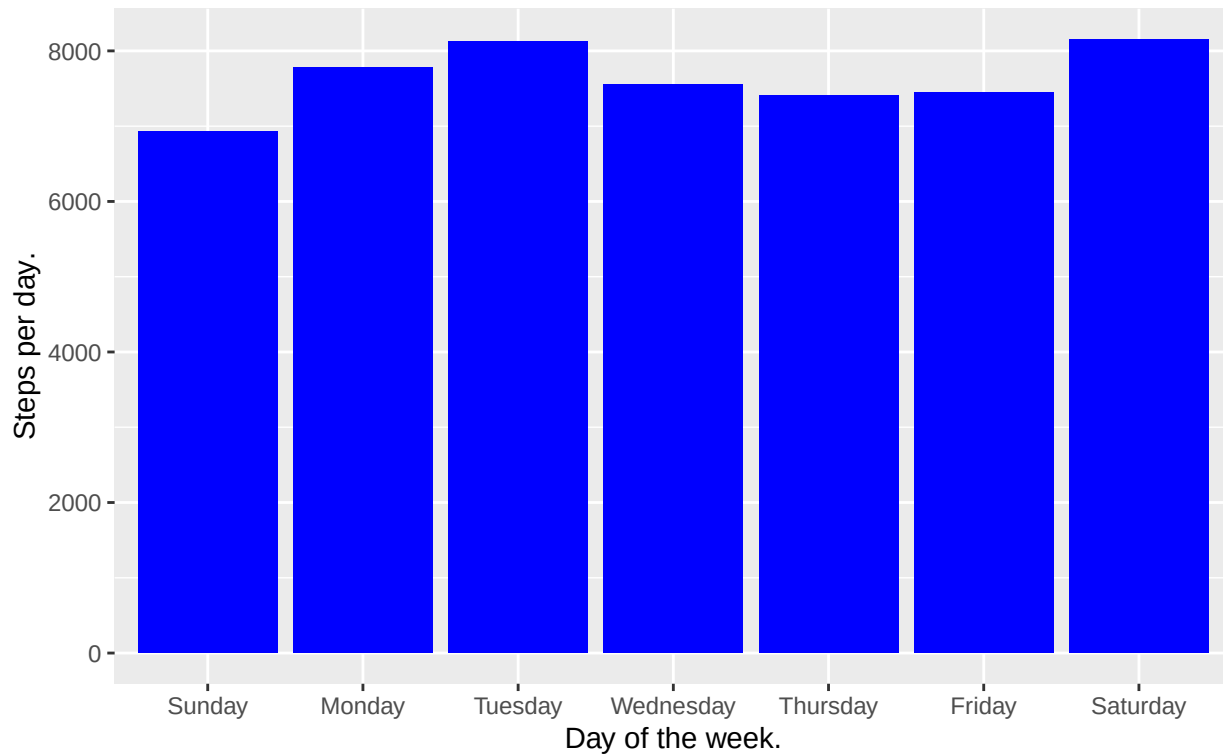
#Viz: Most Active Days by Steps
ggplot(data = daily_activity, aes(x = day, y = AvgTotalSteps)) +
  geom_col(fill = "blue") +
  labs(title = "Most Active Day of the Week by Steps.",
       subtitle = "Shows the days when users took the most steps.",
       x = "Day of the week.",

```

```
y = "Steps per day."
)
```

Most Active Day of the Week by Steps.

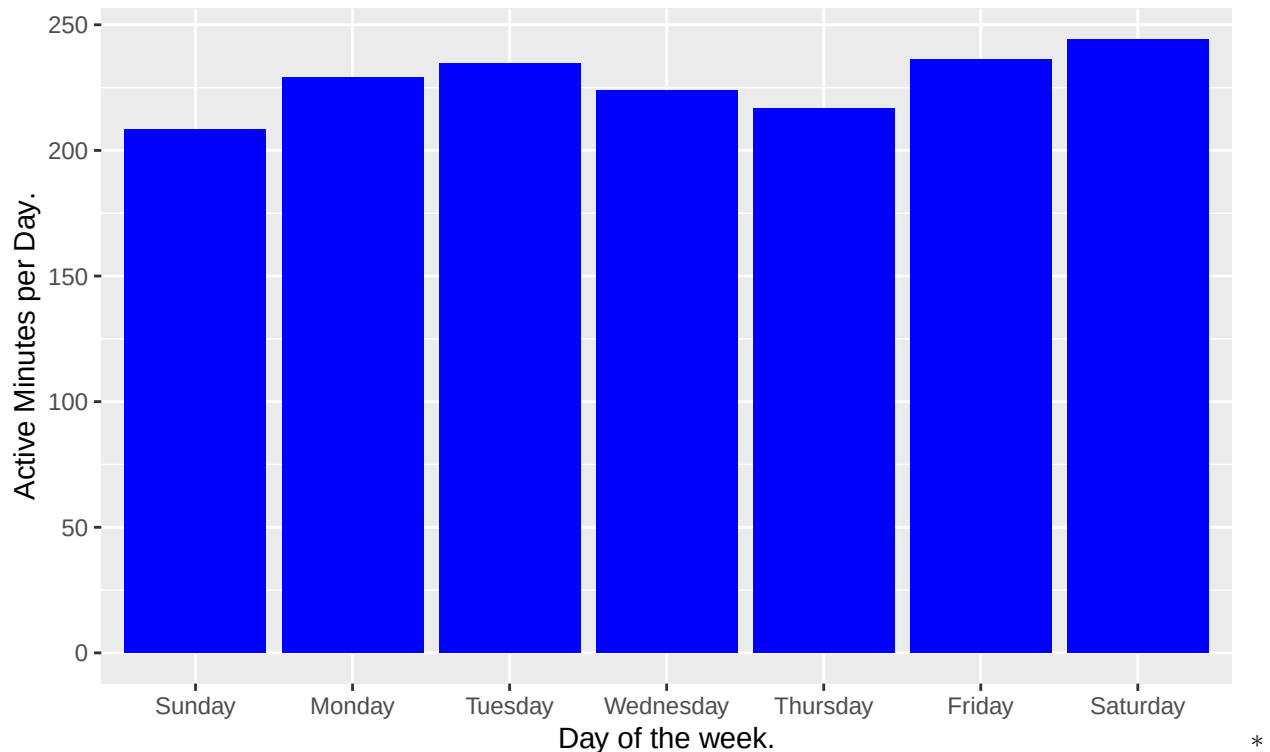
Shows the days when users took the most steps.



```
#Viz: Most Active Days by Active Minutes
ggplot(data = daily_activity, aes(x = day, y = TotalAvgActiveMinutes)) +
  geom_col(fill = "blue") +
  labs(title = "Most Active Day of the Week by Active Minutes.",
       subtitle = "Shows the days when users had the most active minutes.",
       x = "Day of the week.",
       y = "Active Minutes per Day.")
)
```

Most Active Day of the Week by Active Minutes.

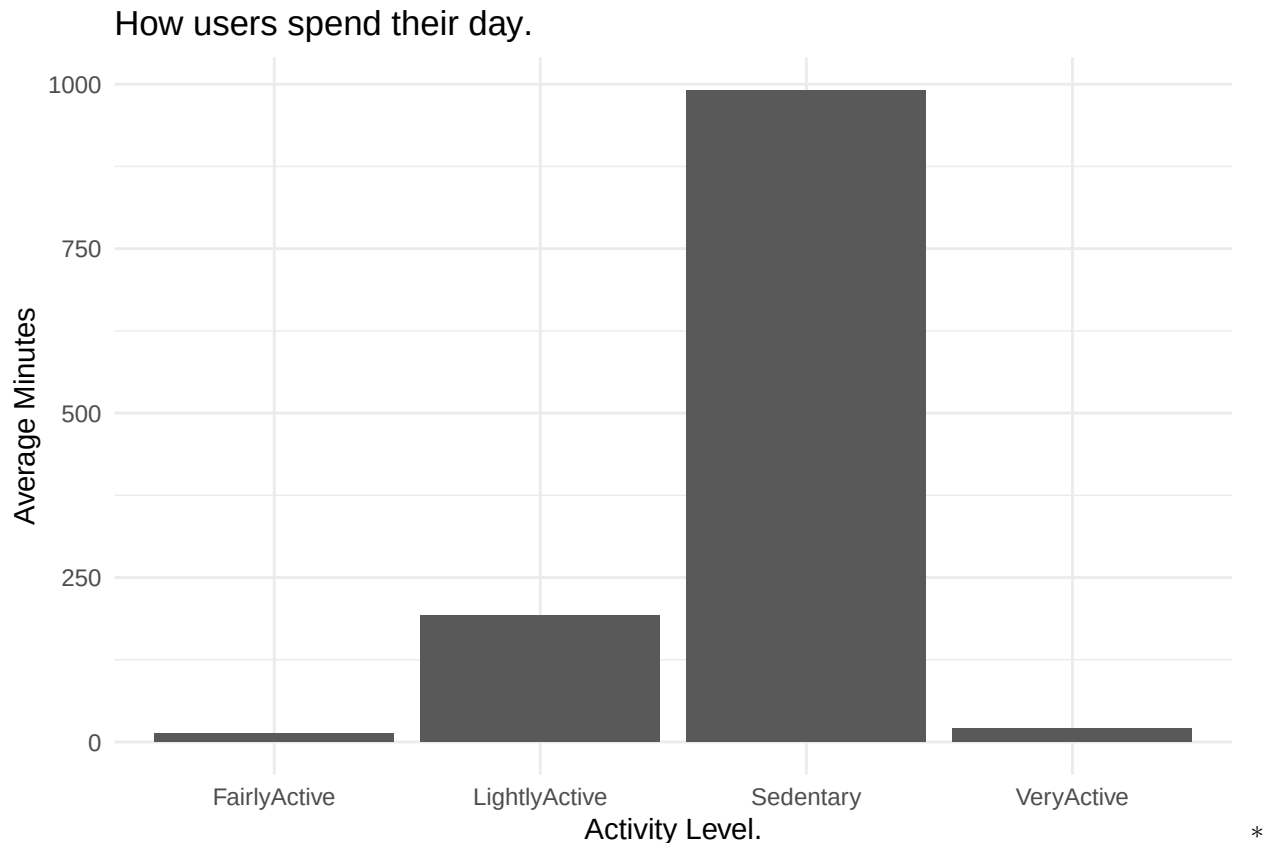
Shows the days when users had the most active minutes.



Insight: We can clearly see from the visualization that Saturday seems to be the most active day of the week for the users, both by steps and active minutes.

```
# How users are spending time on an average day.
activity_composition <- activity %>%
  summarise(
    Sedentary = mean(SedentaryMinutes, na.rm = TRUE),
    LightlyActive = mean(LightlyActiveMinutes, na.rm = TRUE),
    FairlyActive = mean(FairlyActiveMinutes, na.rm = TRUE),
    VeryActive = mean(VeryActiveMinutes, na.rm = TRUE)
  ) %>%
  pivot_longer(
    cols = everything(),
    names_to = "Activity Level",
    values_to = "Average Minutes"
  )

#Viz: Composition of users' days by activity level.
ggplot(data = activity_composition, aes(x = `Activity Level`, y = `Average Minutes`)) +
  geom_col() +
  labs(title = "How users spend their day.",
       x = "Activity Level.",
       y = "Average Minutes"
  ) +
  theme_minimal()
```



Insight: We can clearly see that users spend most of their days being sedentary.

#Merging the Sleep and Activity Datasets.

```
activity <- activity %>%
  mutate(date = as.Date(ActivityDate))

sleep <- sleep %>%
  mutate(date = as.Date(SleepDay))

merged_data <- merge(sleep, activity, by = c('Id', 'date'))
```

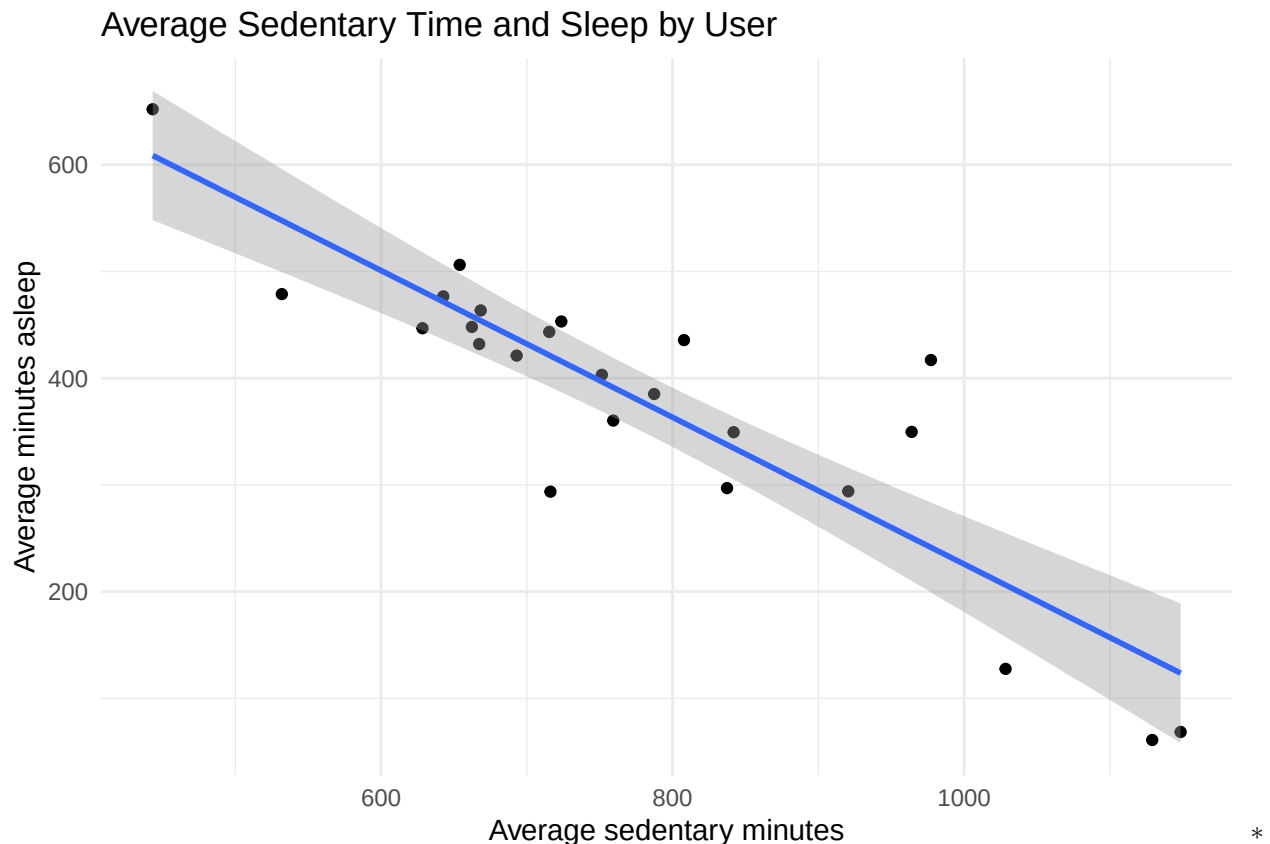
#Checking Users Sedentary Time vs. Time Asleep.

```
user_sleep_activity <- merged_data %>%
  group_by(Id) %>%
  summarise(
    AvgSedentaryMinutes =
      mean(SedentaryMinutes, na.rm = TRUE),
    AvgSleepMinutes =
      mean(TotalMinutesAsleep, na.rm = TRUE),
    .groups = "drop"
  )
```

#Viz: User Sedentary Time vs. Time Asleep.

```
ggplot(user_sleep_activity, aes(x = AvgSedentaryMinutes, y = AvgSleepMinutes)) +
  geom_point() +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "Average Sedentary Time and Sleep by User",
```

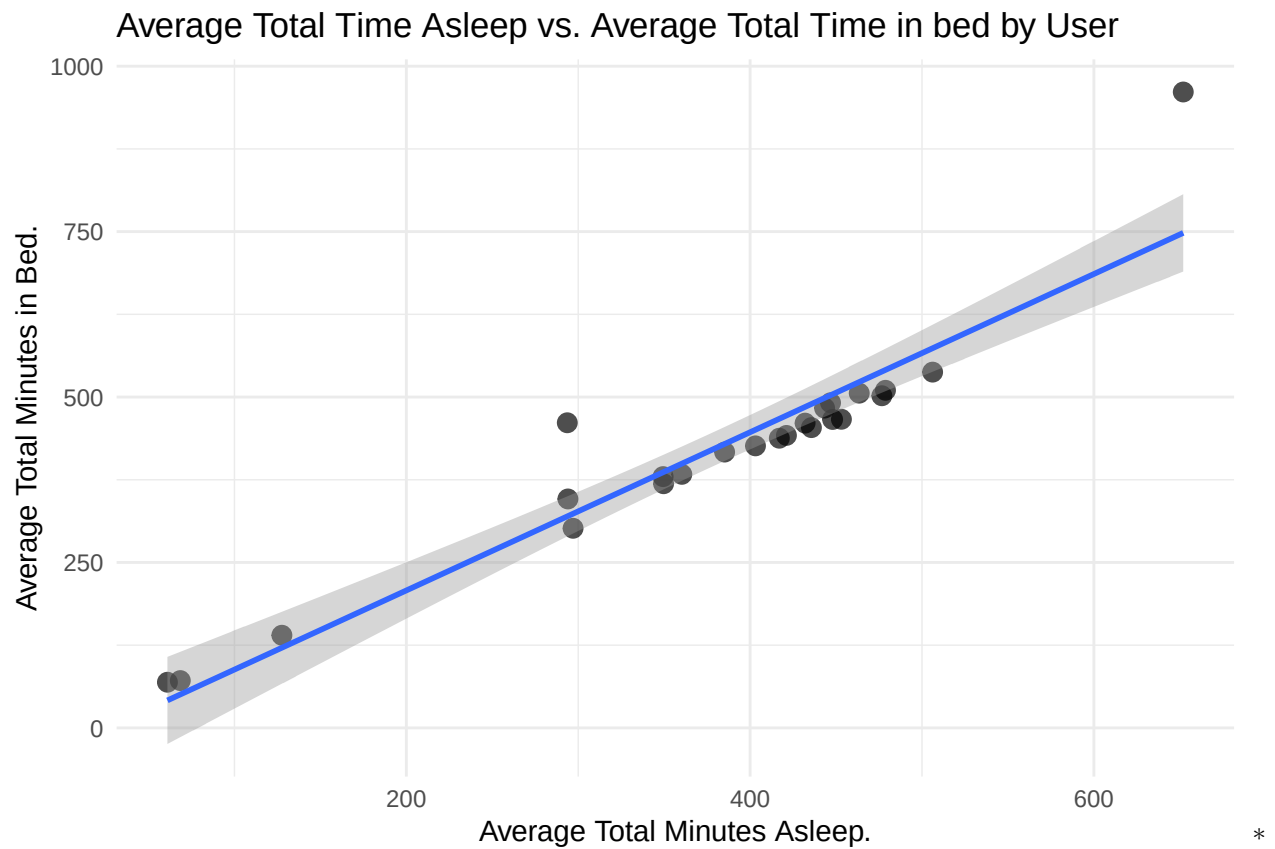
```
x = "Average sedentary minutes",
y = "Average minutes asleep"
) +
theme_minimal()
```



Insight: There seems to be a negative correlation between average sedentary time and time asleep. It means the more time a user spends being sedentary, the less time they sleep.

```
# Total time in bed vs. Total time asleep.
sleep_by_user <- sleep %>%
  group_by(Id) %>%
  summarise(
    AvgTotalMinutesAsleep = mean(TotalMinutesAsleep, na.rm = TRUE),
    AvgTotalTimeInBed = mean(TotalTimeInBed, na.rm = TRUE)
  )

ggplot(sleep_by_user, aes(x = AvgTotalMinutesAsleep,
  y = AvgTotalTimeInBed)) +
  geom_point(alpha = 0.7, size = 3) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "Average Total Time Asleep vs. Average Total Time in bed by User",
    x = "Average Total Minutes Asleep.",
    y = "Average Total Minutes in Bed."
  ) +
  theme_minimal()
```



Insight: It is clearly observed that the more time a user spends in bed, the more time they spend sleeping.

Summary.